

ADVANCED JAVA PROGRAMMING: KEY POINTS

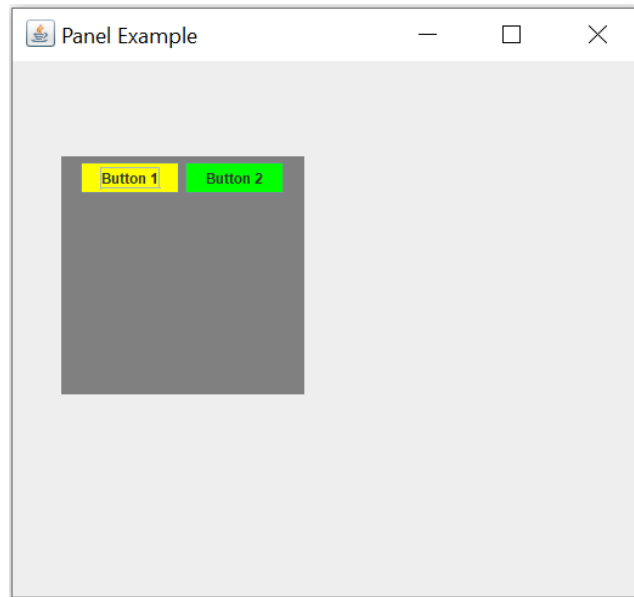
1. Key point 1: JDBC

The screenshot displays the 'University of Kigali - Student Registration' window. The header includes the university logo and the text 'UNIVERSITY OF KIGALI STUDENT REGISTRATION SYSTEM - MYSQL DATABASE'. Below this is a 'Student Registration Form' with a note: 'Fields marked * are mandatory. Data stored in MySQL'. The form contains several fields: 'Full Name *' (filled with 'N. N. Josbert'), 'Gender *' (radio buttons for 'Male' and 'Female', with 'Male' selected), 'Password *' (masked with '*****'), 'Confirm Password *' (empty), 'Date of Birth *' (dropdown for '3' and 'February'), 'Mobile Number *' (filled with '0788547461'), 'Email' (filled with 'josbert@gmail.com'), 'Area *' (dropdown for 'Nyarugenge'), 'State / Province *' (dropdown for 'Kigali City'), and 'Nationality *' (dropdown for 'Rwandan'). A 'Success' dialog box is overlaid on the form, displaying the message: 'Student registered successfully! Welcome, N. N. Josbert!' with an 'OK' button. At the bottom of the form are 'Clear' and 'Submit' buttons. The footer of the window reads 'University of Kigali - Student Registration System - Java Swing + MySQL (JDBC)'.

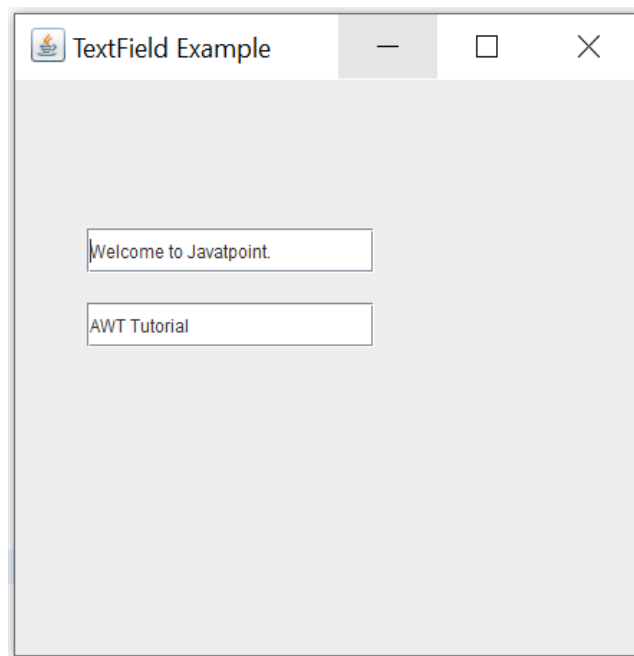
2. Key point 2: JDBC

The screenshot shows the 'Kigali Employee Data Entry' window. The title bar reads 'Kigali Employee Data Entry'. The main heading is 'KIGALI EMPLOYEE DATA ENTRY FORM'. The form contains the following fields: 'Company Name:' (text input), 'Industry:' (dropdown menu with '--Select--'), 'Employee Names:' (text input), 'Gender:' (dropdown menu with '--Select--'), 'Post:' (dropdown menu with '--Select--'), 'Min Salary:' (dropdown menu with '--Select--'), and 'Max Salary:' (dropdown menu with '--Select--'). At the bottom of the form is a green 'SAVE DATA' button.

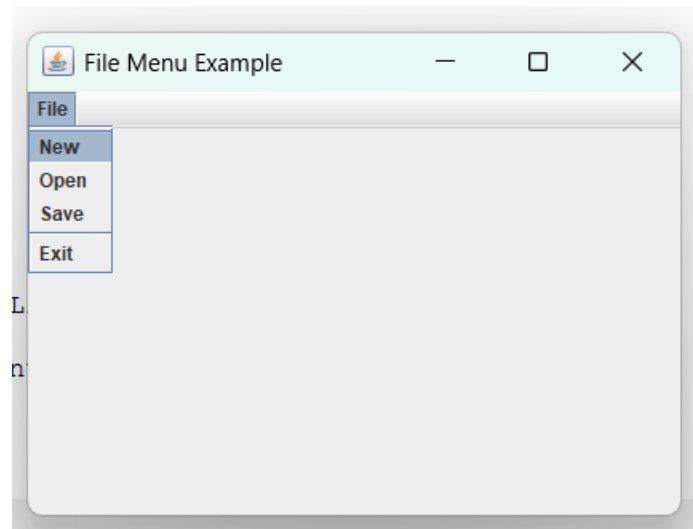
3. Key point 3: GUI



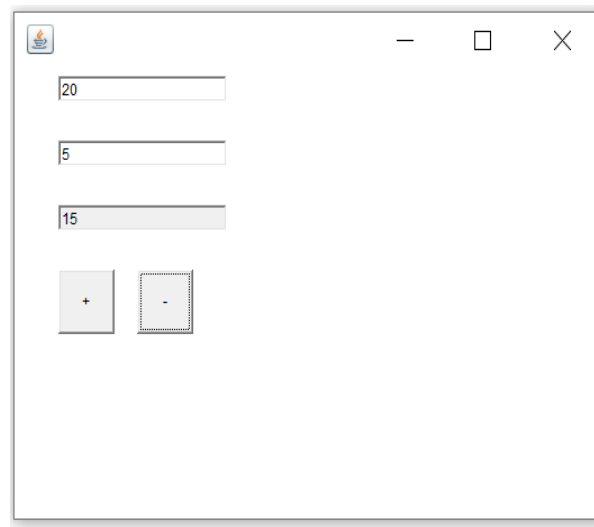
4. Key point 4: GUI



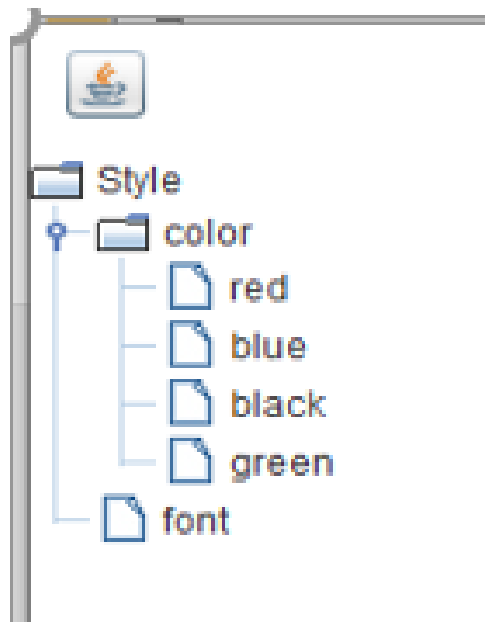
5. Key point 5: GUI



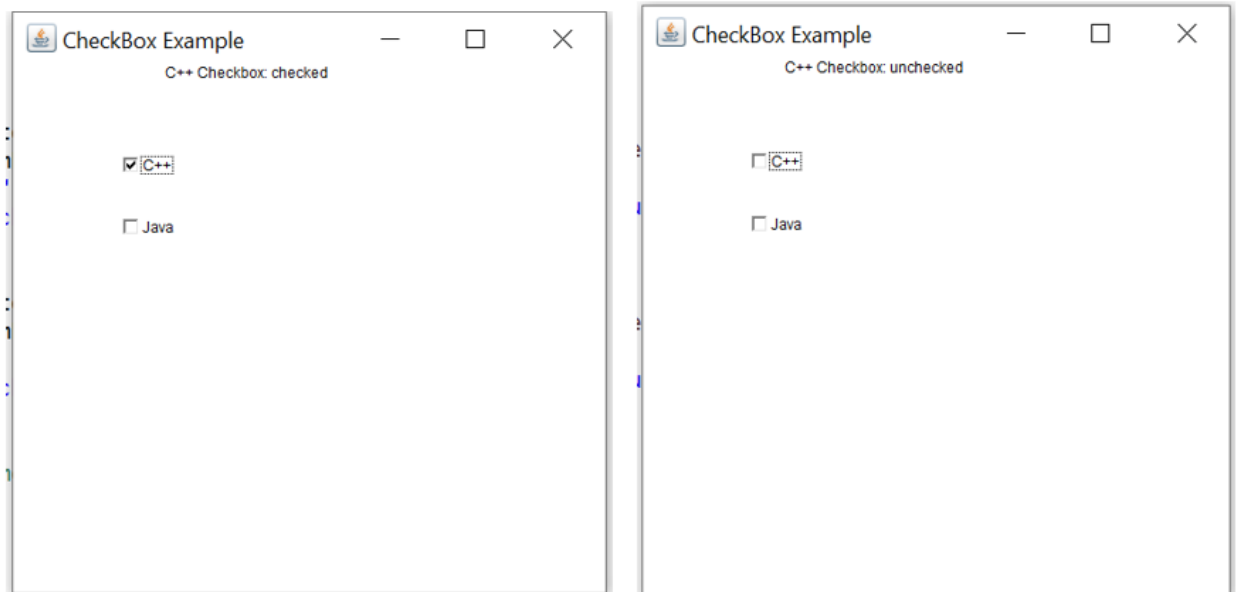
6. Key point 6: GUI



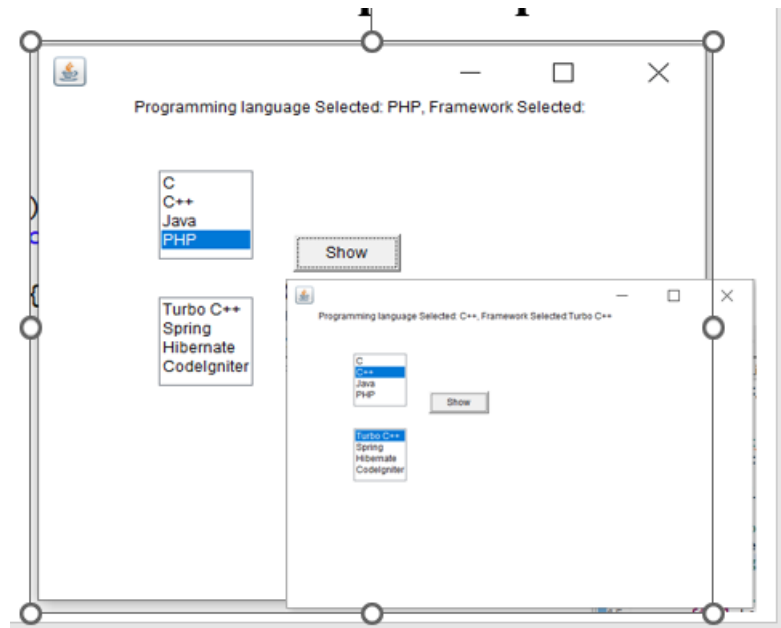
7. Key point 7: GUI



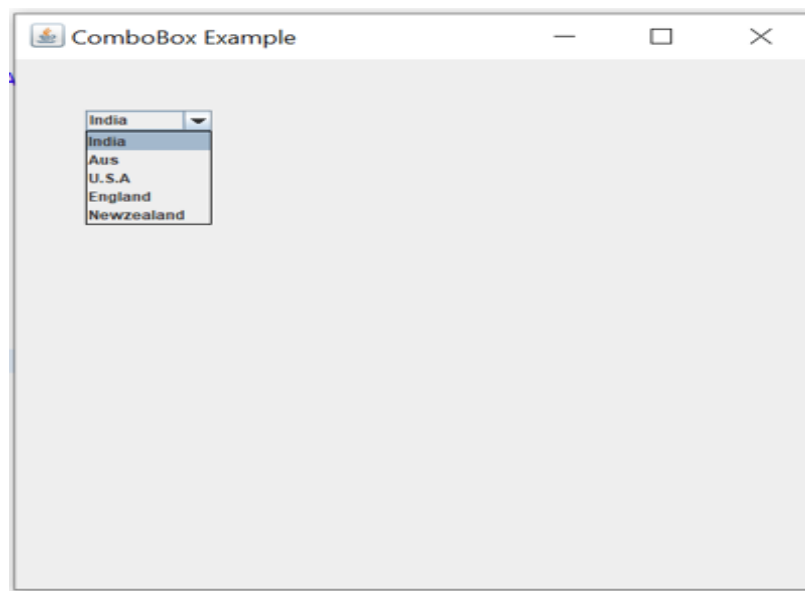
8. Key point 8: GUI



9. Key point 9: GUI



10. Key point 10: GUI



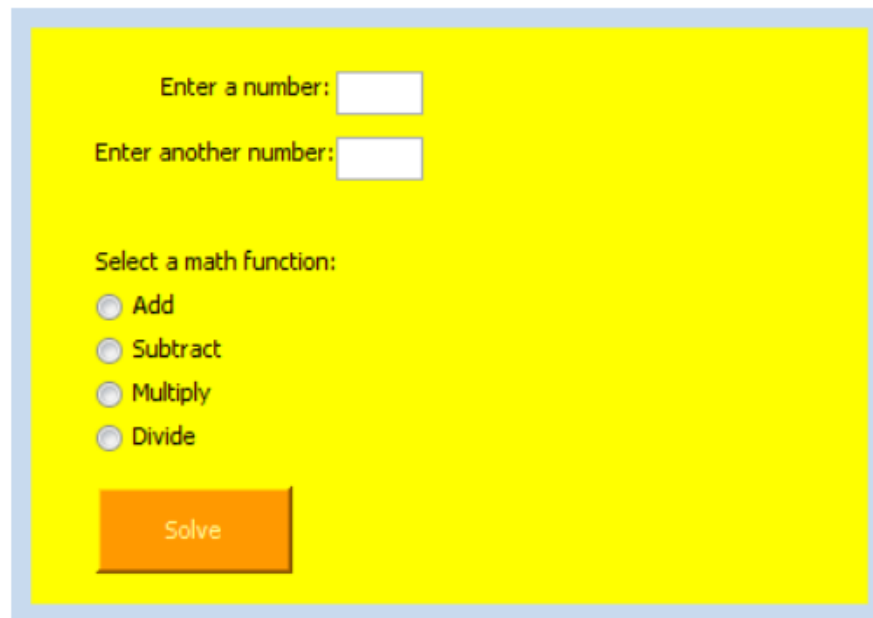
11. Key point 11: Understand Theory:

Examine how the Java compiler, bytecode, and Java Virtual Machine (JVM) work together to enable platform independence in Java applications. Offer a detailed description showing how these components allow a program written once to run on different operating systems without modification, and analyze how this mechanism improves portability and reliability when compared with traditional machine dependent programming languages.

12 . Key point 12: Understand Theory:

Assess the appropriateness of the Java execution environment for developing large scale systems. In your discussion, consider factors such as system performance, security, cross platform compatibility, and ease of maintenance. Explain the advantages of using Java for such a system and also discuss potential limitations or challenges that developers might encounter when implementing Java in a large distributed environment.

13. Key point 13: NetBeans FORM GUI



Enter a number:

Enter another number:

Select a math function:

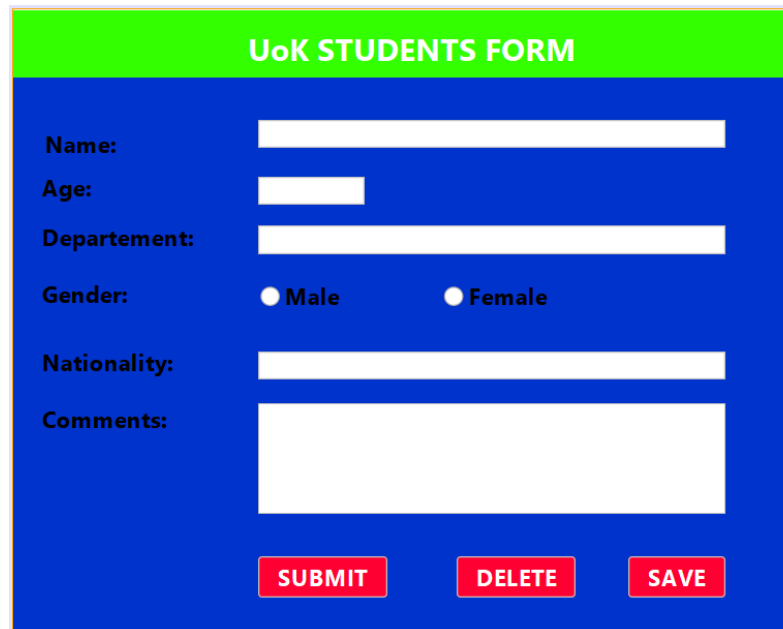
☐ Add

☐ Subtract

☐ Multiply

☐ Divide

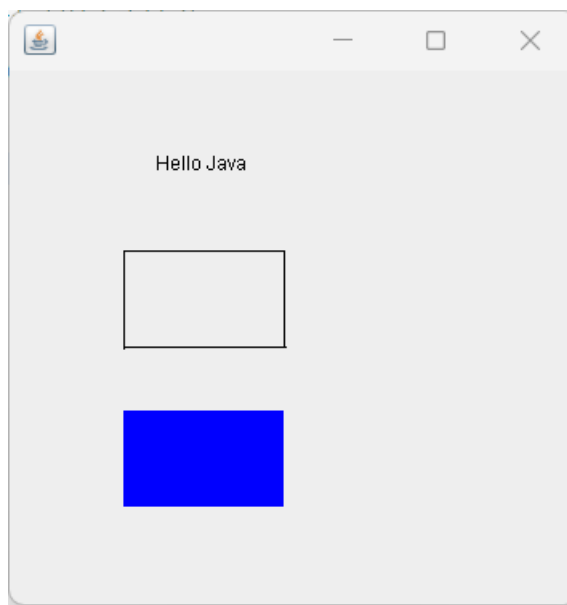
14. Key point 14: NetBeans FORM GUI



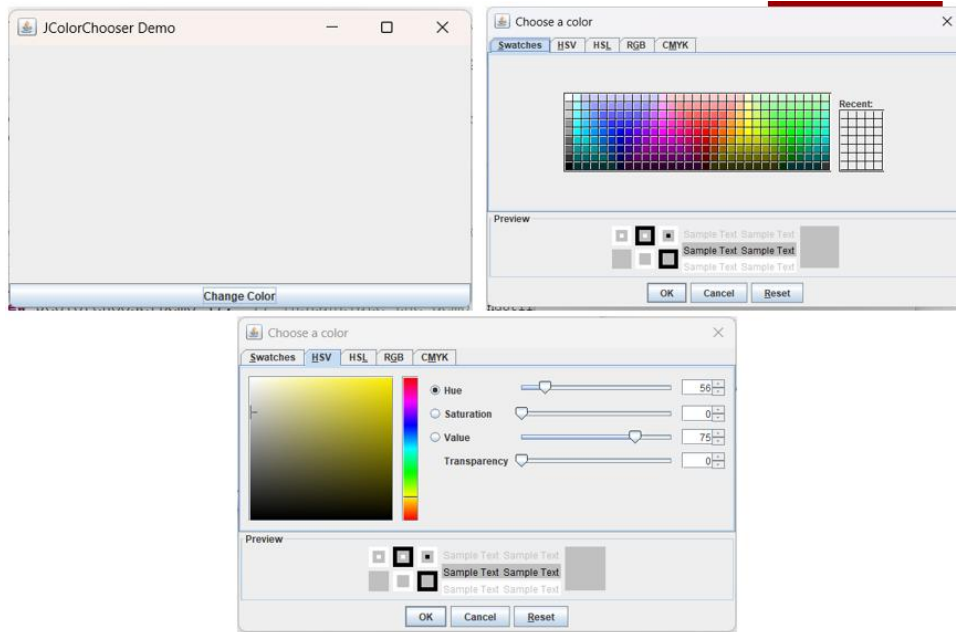
A NetBeans Swing window titled "UoK STUDENTS FORM" with a green title bar. The window has a blue background and contains the following fields and controls:

- Name:** A single-line text input field.
- Age:** A single-line text input field.
- Departement:** A single-line text input field.
- Gender:** Two radio buttons labeled "Male" and "Female".
- Nationality:** A single-line text input field.
- Comments:** A multi-line text area.
- Buttons:** Three red buttons labeled "SUBMIT", "DELETE", and "SAVE" are positioned at the bottom right.

15. Key point 15: 2D Shapes and colors



16. Key point 16: 2D Shapes and colors



17. Key point 17: Colors

Color constant	Color	RGB value
<code>public final static Color ORANGE</code>	orange	255, 200, 0
<code>public final static Color PINK</code>	pink	255, 175, 175
<code>public final static Color CYAN</code>	cyan	0, 255, 255
<code>public final static Color MAGENTA</code>	magenta	255, 0, 255
<code>public final static Color YELLOW</code>	yellow	255, 255, 0
<code>public final static Color BLACK</code>	black	0, 0, 0
<code>public final static Color WHITE</code>	white	255, 255, 255
<code>public final static Color GRAY</code>	gray	128, 128, 128
<code>public final static Color LIGHT_GRAY</code>	light gray	192, 192, 192
<code>public final static Color DARK_GRAY</code>	dark gray	64, 64, 64
<code>public final static Color RED</code>	red	255, 0, 0
<code>public final static Color GREEN</code>	green	0, 255, 0
<code>public final static Color BLUE</code>	blue	0, 0, 255

18. Key point 18: 2D Shapes and colors



19. Key point 19: JavaBeans

Write a Java class named Student that follows JavaBeans conventions. The class should include a private property called name to store the student's name. It should provide a no-argument constructor, along with methods that allow setting and retrieving the value of the name property. The class must also support object serialization so that its objects can be saved to a file.

20. Key point 20: JavaBeans

by adding a main application class named MainApp that demonstrates how a Student object can be saved to a file and later loaded back into the program. The program should first create a Student object and assign a name to it. It should then save the object into a file using object serialization. After saving the object, the program should read the same object back from the file using deserialization and display the student's name on the screen. During execution, the program must print confirmation messages. When the object is successfully saved, it should display "Object has been serialized (saved)." When the object is successfully loaded, it should display "Object has been deserialized (loaded)." Finally, it should display the student's name in the format "Student Name: Mugabo Mohammed".

21. Theory related to JavaBeans and BeanInfo.

22. Key Points 22: GUI

